

HackerRank Algorithmic Problem-Solving & Portfolio

3rd Semester Portfolio Report

Name	Mitarth Rai	Roll Number	R25EJ073
University	REVA University	Branch	CSIT
Semester	3	Section	B
Language	C	HackerRank	raimitarth_vns

1. Project Overview

This portfolio contains five HackerRank problem-solving programs completed in C for the 3rd Semester portfolio activity. The solutions cover arrays, strings, comparisons, dynamic data structures, and frequency-based string processing. Each solution was submitted to HackerRank and the accepted submission screenshots are included as evidence.

2. HackerRank Problems

Problem	Language	Status	Complexity
Diagonal Difference	C	Accepted	$O(N) / O(1)$
Dynamic Array	C	Accepted	$O(N+Q) / O(N)$
Time Conversion	C	Accepted	$O(1) / O(1)$
Compare the Triplets	C	Accepted	$O(1) / O(1)$
Sparse Arrays	C	Accepted	$O(N+Q) / O(N)$

3. GitHub Repository

Repository: <https://github.com/Mitachi12/HackerRank-3rdSem-Portfolio>

GitHub Profile: <https://github.com/Mitachi12>

HackerRank Profile: https://www.hackerrank.com/profile/raimitarth_vns

4. Repository Structure

The repository is organized into separate folders for each HackerRank problem. A dedicated HackerRank-Evidence folder contains screenshots of the accepted submissions. The repository also contains the reflection and this studio report.

```
HackerRank-3rdSem-Portfolio/  
■■■ 01-Diagonal-Difference/solution.c  
■■■ 02-Dynamic-Array/solution.c  
■■■ 03-Time-Conversion/solution.c  
■■■ 04-Compare-the-Triplets/solution.c  
■■■ 05-Sparse-Arrays/solution.c  
■■■ HackerRank-Evidence/  
■■■ reflection.md  
■■■ Studio-Report-2-Page-Final.pdf
```

5. Reflection

Working on these HackerRank problems improved my understanding of basic programming and problem-solving concepts in C. The problems gave me practice with arrays, strings, comparisons, dynamic memory allocation, and data structures. Submitting the solutions to HackerRank also helped me test the programs against multiple test cases and pay attention to input handling and edge cases. Using C provided additional practice with memory allocation and pointers, especially for Dynamic Array and Sparse Arrays. I also learned how to organize solutions into separate folders and maintain the work in a GitHub repository. Overall, the activity improved my confidence in solving basic algorithmic problems and writing organized C programs.

6. Evidence

The HackerRank-Evidence folder contains screenshots showing Accepted submissions for all five required problems. The repository links and submission evidence are included so the completed work can be reviewed.

Prepared by: Mitarth Rai | R25EJ073